

Advanced Statistical Data Analysis

Worksheet Week 4

Exercise 1: Nelson (1982, pp. 407–409) discusses an experiment to determine the relationship of time in use to the number of fissures that develop in turbine wheels. The dataset given in `turbines.dat` contains three columns: operation time of the turbine (`Hours`), number of turbine wheels in operations (`Turbines`), and number of turbine wheels with fissures (`Fissures`).

- Generate a graph that shows the operation time and the response variable “defective” turbine wheels. What kind of distribution can be used to describe the response variable?
- Fit a logistic regression model with response variable “defective” turbine wheels and explanatory variable operation time (`Hours`). Write down the model equation in mathematically correct notation.

Remember that you must specify a pair of numbers identifying the number of “successes” and the number of “failures” on the right hand side of “ $\sim$ ”. (Be careful what is meant by “successes” and “failures” in this example!)

- Describe the relationship between operation time and the probability of “defective” turbine wheels using the estimated coefficients of the logistic regression model.
- What is the estimated probability of a “defective” turbine wheel for an operation time of the turbine of 3'000 hours?
- Overlay the fitted curve into the graphic of part (a). Based on the graphic, estimate the operation times at which 10%, 30%, 50%, 70% and 90% of the turbine wheels exhibit any fissures.
- Repeat this analysis using the probit and the clog-log link. Discuss the three analyses. (Are there any differences?)

R hints:

- Display fitted curve

```
> turbN <- data.frame(Hours=seq(100, 10000, by=100))
> turb.p <- predict(turb.logit, newdata=turbN, type="response")
> lines(turbN$Hours, turb.p)
> abline(h=c(0.1,0.5, 0.7, 0.9), lty=7, col=2)
```

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Exercise 2: The dataset `creditdata.csv` contains data from a German bank. 932 random customers were selected and analysed to see whether they were able to repay their loan on time. The following information about the customers is also known

Creditability	loan repaid late or not repaid (0) or repaid on time (1)
Duration	Credit duration in months
Credit.Amount	Amount of the credit
Account.Balance	Account balance at this bank ($1 \leq 0$ €, $2 = 0-200$ €, $3 = \geq 200$ €, $4 = \text{no account}$)
Savings	savings at this bank ($1 = \leq 100$ €, $2 = 100-500$ €, $3 = 500-1000$ €, $4 = \geq 1000$ €)
NCredits	number of loans with this bank
Gender	gender of the customer (f = female, m = male)
Marital.status	marital status of the customer (single, married, divorced)
Age	Age of the customer (in years)
Occupation	Job status of the customer (unemployed, employed, self-employed)

Our aim is to model the individual credit worthiness of this bank's customers. The customers can be assumed to be independent of each other.

- Fit a logistic regression model with response variable **Creditability** and the two explanatory variables **Account.Balance** and **Duration**. Please note that **Account.Balance** is a categorical variable. What do the estimated coefficients tell you?
- Plot the estimated model graphically. Does the graphical representation confirm your interpretation?
- Using the graph in (b), estimate the terms for which more than 50% of customers would repay their loan.

Exercise 3: (Optional) Let us model the probability of surviving a premature birth. The explanatory variable which is probably most important is the weight at birth. If we categorise the variable weight we can summarize our data in a table given in the file `birth-weight.dat`.

weight	550	650	750	850	950	1050	1150	1250	1350	1450
m	10	14	27	22	32	28	22	26	34	32
Y	0	2	9	8	23	21	19	19	31	29

- Plot the data using circles or sunflowers as symbols in a scatterplot proportion against weight. Plot the data in a scatterplot (empirical) logit-transformed proportions against weight. Can we improve by an additional transformation such that the relationship shows up more adequate.
- Fit the model using the least squares approach with empirical logits. Plot the resulting regression line in the previous plot.
- Compare the result of part (b) with the result of glm fit by displaying both fitted lines in the scatterplot proportion against weight.